How Machine Learning Works

Lecture 3 — models, data and the ML workflow



Today’s plan



Session 1

What is a model?

- From learning agents to ML
- A model is a function
- Features, labels, examples
- Regression vs. classification

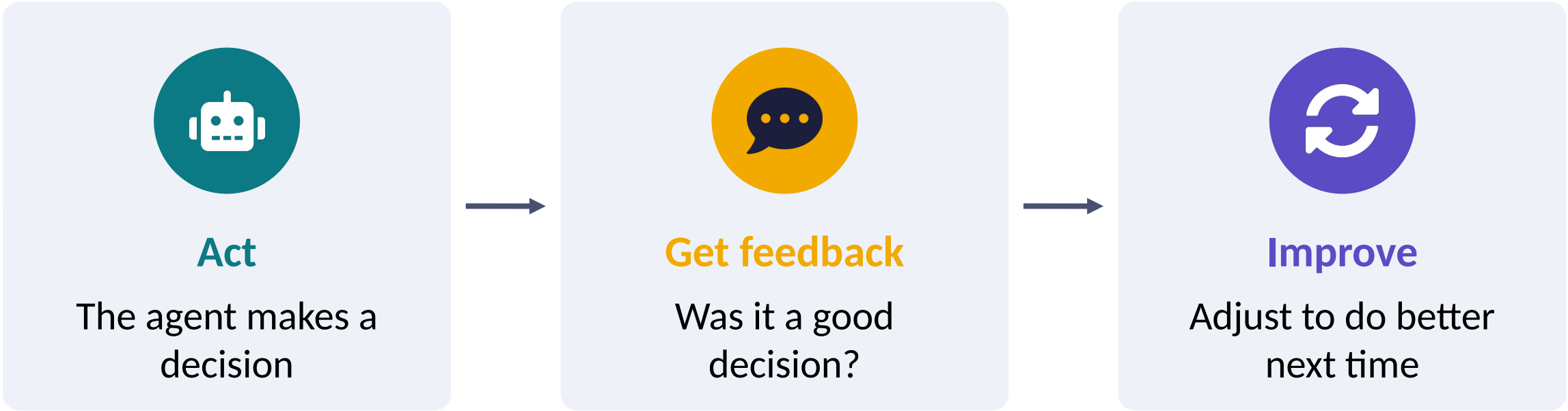


Session 2

The ML workflow

- Seven steps, start to finish
- Why we hide test data
- Overfitting and underfitting
- How ML goes wrong

Last week: the learning agent



Machine learning = the part that turns experience (data) into better decisions.

Why learn instead of writing rules?



Too many rules

Try writing rules to recognise a face in a photo



The world changes

Fraudsters change tactics; hand-written rules go stale



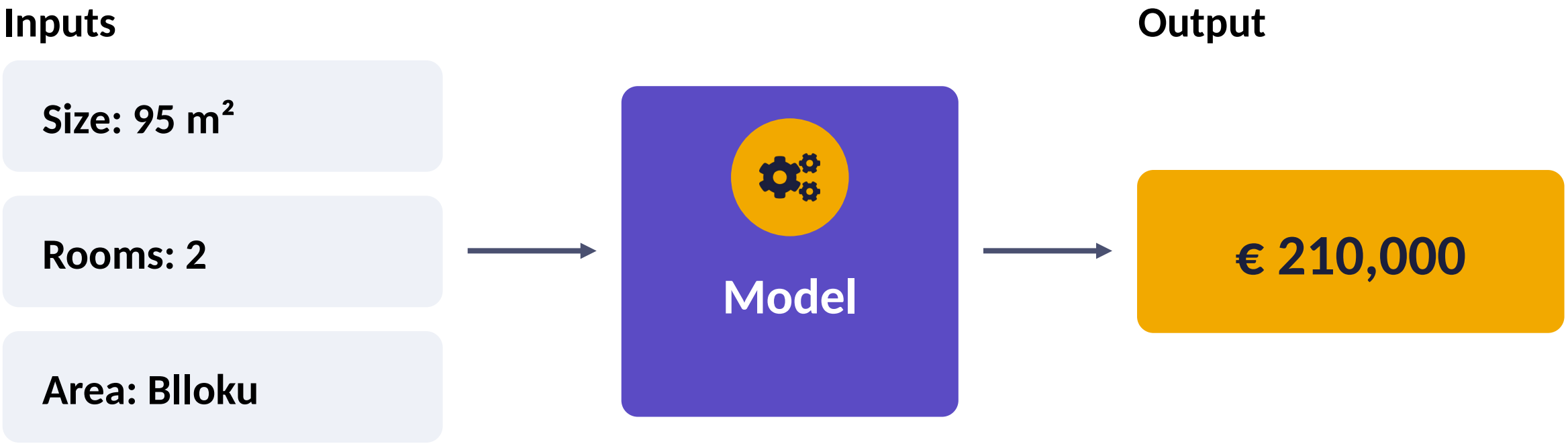
We don’t know the rules

Why does a customer buy? The data may know better

What is a model?

The single idea behind every ML system

A model is a mathematical function



Like an Excel formula — but nobody writes it. It is learned from examples.

The vocabulary: features, label, example

Size (m^2)	Rooms	Area	Price (€)
95	2	Blloku	**210,000**
70	2	Kombinat	**85,000**
110	3	Astir	**140,000**
60	1	Don Bosko	**70,000**

Features

the inputs (columns we know)

Label

what we want to predict

Example

one row = one house

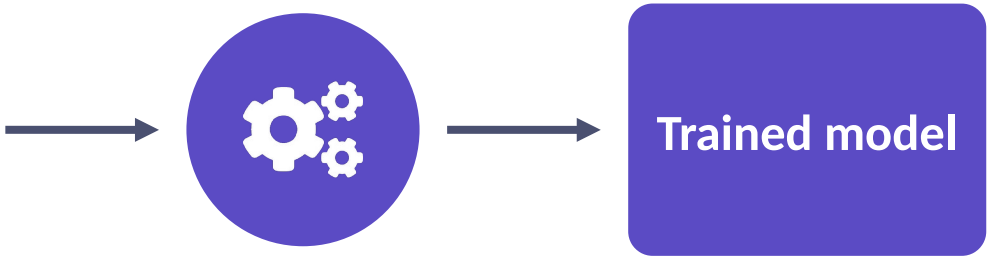
In a spreadsheet: features are the columns you know, the label is the column you want to fill in.

Two phases: training and prediction

1. Training

`model.fit(...)`

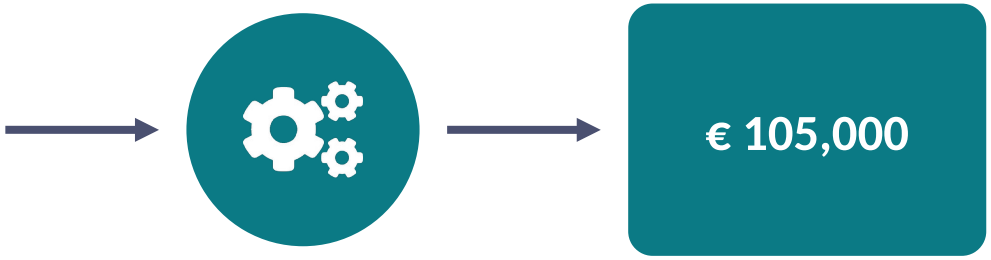
Many past houses — with their prices



2. Prediction

`model.predict(...)`

A new house: 80 m², 2 rooms, Astir, price = ?



The model only learns once. Then it can predict as many new cases as we like.

Two kinds of questions



Regression

How much? How many?

The answer is a number

- Price of a house
- Next month’s sales
- Delivery time in days



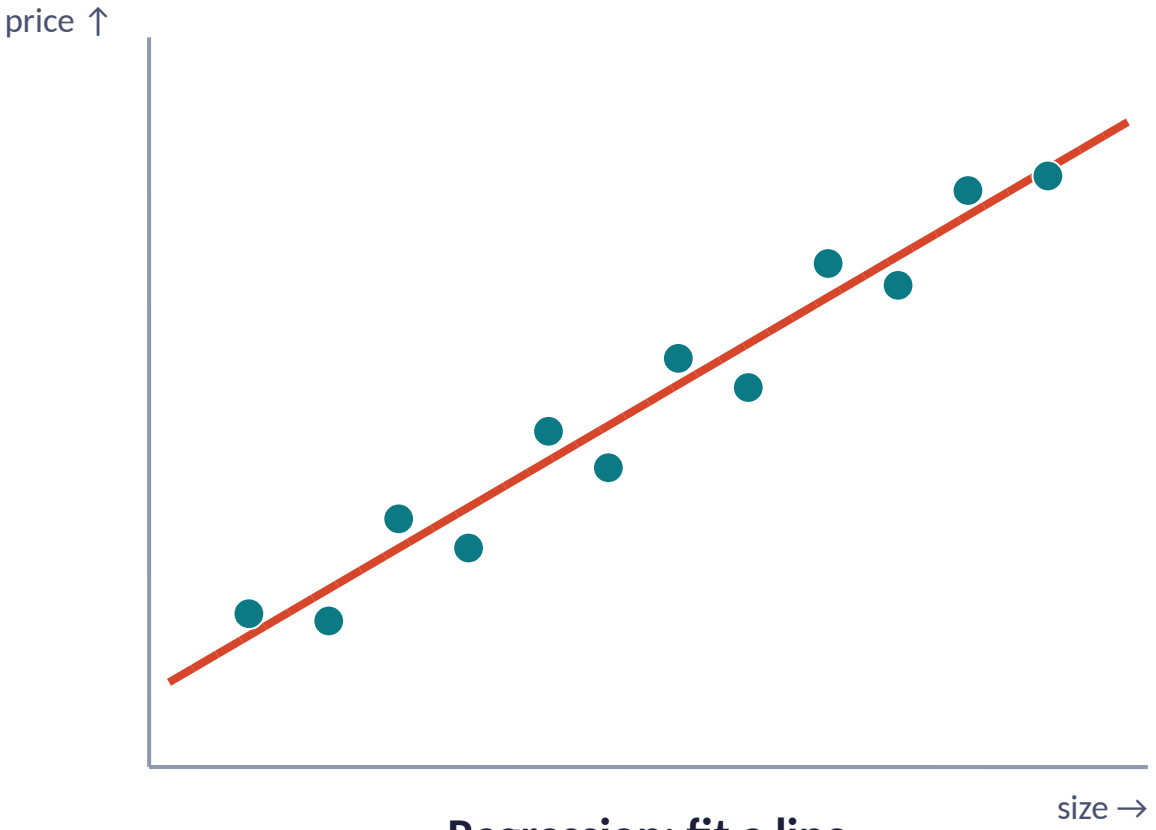
Classification

Which one? Yes or no?

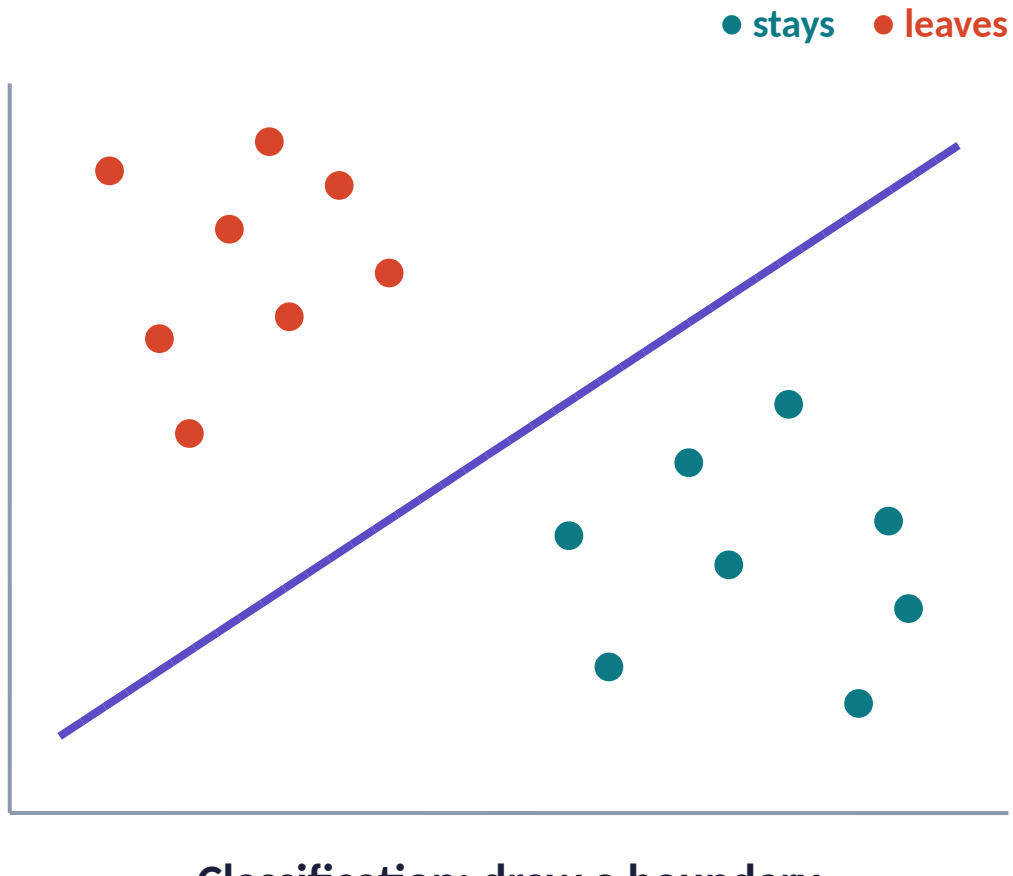
The answer is a category

- Will this customer leave?
- Is this email spam?
- Which product category?

What the model actually learns



Regression: fit a line



Classification: draw a boundary

Illustration — not real data

Quick check: regression or classification?

1

How many units will we sell next week?

2

Will this loan be repaid?

3

How long will this customer stay with us?

4

Which department should handle this complaint?

5

What salary should we offer?

6

Will this transaction be over the fraud limit?

Where do the labels come from?

Past records	Last year's sales, past loans that were repaid or not	**Cheap — the answer is already there**
People label them	Doctors mark X-rays, staff tag complaints	**Expensive and slow**
User actions	Clicks, purchases, "report spam" buttons	**Free but noisy**

Frame your own ML problem

In pairs

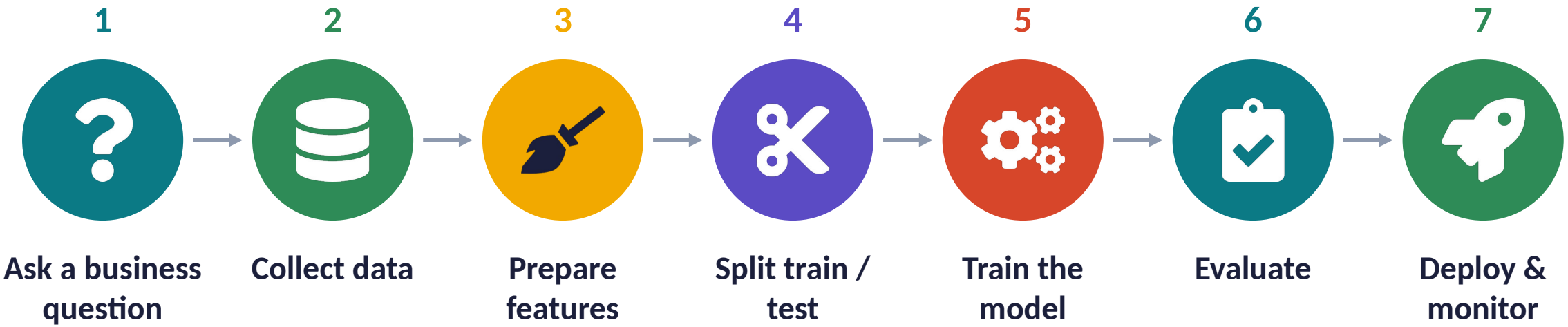
Pick a business you know (a bank, a shop, a hotel…). Find one thing worth predicting.

1. What is the label — what do you want to predict?
2. What are the features — what would you know in advance?
3. Regression or classification?

The ML workflow

From a business question to a working model

Seven steps of every ML project



It is a loop. Customers, prices and markets change — a deployed model must be watched and retrained.

Step 1 matters most: a precise question



Vague

“Use AI to improve our sales.”

- Predict what?
- For whom? By when?
- What do we do with it?



Precise

“Predict which customers will cancel next month, so we can call them with an offer.”

- Label: cancels — yes/no
- Timing: next month
- Action: retention call

Garbage in, garbage out (GIGO)



Missing

Empty cells: no age, no income



Wrong

Typos, age = 250, duplicates



Outdated

Pre-pandemic shopping habits



Unrepresentative

Only city customers, but we sell nationwide

In real projects, most of the time goes into the data — not the model.

Why we hide some data: the train/test split



the model learns from this

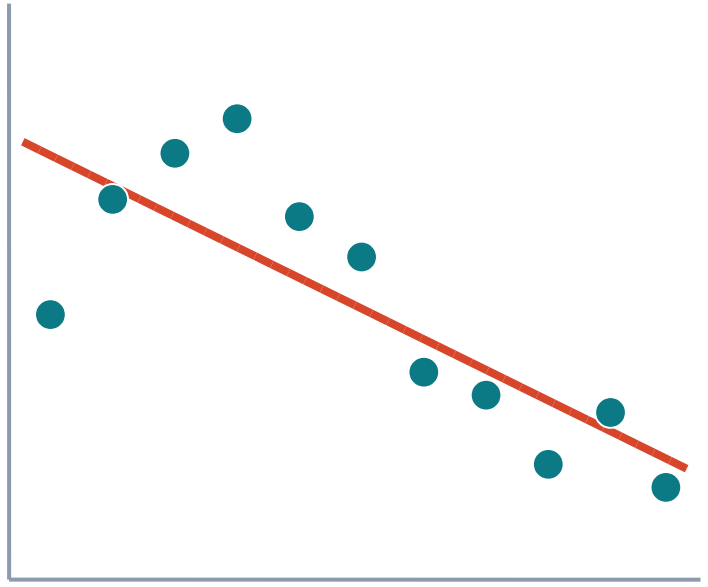
hidden until the end



The exam analogy

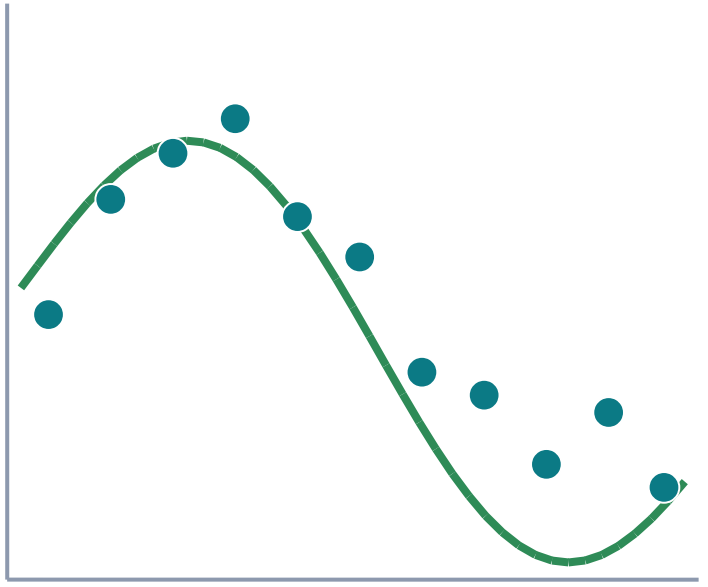
A student who memorises last year’s exam scores 100% on it — and fails a new one. We test on questions the model has never seen.

Underfitting and overfitting



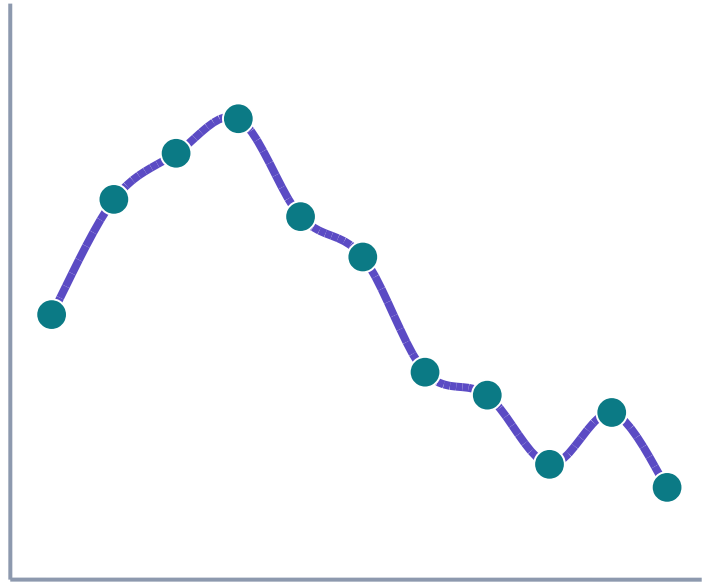
Too simple

Underfitting



Just right

Good fit



Too complex

Overfitting

Overfitting = memorising the training data, including its noise.

How do we measure success?

Regression

Average error

€12,000

Our price predictions are off by €12,000 on average

Classification

Accuracy

90%

The model is right on 90 of every 100 customers

Always measured on the test data — never on the data the model learned from.

Accuracy can lie

A fraud model that always answers “not fraud”:

99%

accuracy

0

frauds caught

When the thing we care about is rare, accuracy alone tells us almost nothing.

Four ways ML goes wrong



Biased data

A hiring model learns past bias — and repeats it



Leakage

Using information you won’t have when predicting



Wrong target

Optimising clicks instead of satisfaction (King Midas)



Drift

The world changes; the model doesn’t

Class exercise: frame these problems

Problem	Label?	Features?	Type?	Measure?
Telecom: which customers will cancel?				
Supermarket: how much milk to order next week?				
Bank: will this loan be repaid?				

In the seminar: the whole workflow in 10 lines

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

data = pd.read_csv("houses.csv")
X, y = data[["size", "rooms"]], data["price"]
X_tr, X_te, y_tr, y_te = train_test_split(X, y)
model = LinearRegression()
model.fit(X_tr, y_tr)
model.score(X_te, y_te)
```

- Load data, pick features & label
- Split train / test
- Train the model
- Evaluate on test data

Key takeaways

- A model is a function learned from examples
- Features are the inputs, the label is what we predict
- Number → regression; category → classification
- Always test on data the model has never seen
- Most ML problems start — and fail — with the question and the data

Next week: Data as representation — tables, vectors, matrices. Seminar: NumPy.